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Accelerating Molecular Generation through Representation Alignment

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Abstract

Write a summary of the whole thing. Make sure it fits on one page.

Contents

Chapter 1

Introduction

The evolution of generative machine learning in scientific domains is defined by the shifting tensions between data scale, data quality, and structural priors. Our report examines a recent addition to this story, *representation alignment*, in two regimes for de novo molecular generation — small molecules and protein backbones — and characterises its behaviour. In this introduction, we trace the narrative arc of research trends in molecular generation, and highlight the motivation for our work: the emerging need for data-efficient modelling in a world of prior-light models.

1.1 A history of trade-offs

The dominant ideas in generative modelling are *denoising diffusion* and its offshoot *flow matching* [1, 2], which found breakthrough success in the realm of image generation. This paradigm has since been adapted for de novo design challenges across scientific fields, such as generating protein backbones [18, 19, 14], inorganic materials [21], and drug-like small molecules [15]. Training these models is typically expensive in both data and compute, with state-of-the-art image generators routinely training on billions of examples [32]. However, the corpus of available high-quality molecular data sits orders of magnitude below this. Experimental determination is slow and expensive, which means datasets are correspondingly small. For example, protein structure generators have largely been trained on at most half a million structures [14], which may explain why they have not yet matched the maturity of their vision-domain counterparts.

Historically, researchers compensated for the small scale of high-quality molecular data by encoding structural priors directly into model architectures [38]. SE(3)-equivariant architectures, for example, enforce the rotational and translational symmetries expected of molecules by construction [40]. But hard structural priors come with a computational cost, and the operations required to encode geometry are a bottleneck to scaling models.

A recent wave of research has moved entirely the other way: drop hard priors, use bare-bones non-equivariant transformer stacks, and rely on data scale and augmentation to teach the model what equivariance buys. This shift fits a broader trend in machine

learning on leveraging representation instead of model architecture to inject soft inductive biases for structured data. Vision Transformers [41] showed that the spatial-locality and translation-equivariance priors of convolutional neural nets (CNNs) can be learned rather than imposed, given sufficient scale. Graph Transformers like TokenGT [42] have made the same case for graphs. AlphaFold 3 [44] stripped many of the SE(3)-equivariant biases of AlphaFold 2 [43], replacing the structure module with a diffusion model operating on raw coordinates. Brehmer et al. [45] make the empirical scaling case directly: more data through a simpler model beats less data through a richer one. In our work, we examine Tabasco [15] and Proteina [14], which are flow-based generators built on largely vanilla transformers for small molecules and protein backbones respectively.

Given the difficulty of procuring data in scientific domains, data scale must necessarily come from non-experimental sources, which may be of lower quality. For example, recent efforts in protein generation have leaned on synthetic structures from folding-model predictions [14, 34] to expand training datasets into the tens of millions. However, even these enhanced corpora are arguably insufficient. Metagenomic databases catalogue over two billion natural protein sequences [36], and the theoretical sequence space at a median protein length [37] is hundreds of orders of magnitude greater still. Even at the frontier of the field's data scale, we sample but a sliver of the distribution we wish to model.

In summary, the field has traded hard structural priors for architectural simplicity, and curated data for synthetic scale. However, this trade-off dilutes the structural signal available to a model. Furthermore, even our best attempts at creating large datasets may not be large enough. These twin challenges amplify the need for techniques that make training still more efficient without sacrificing quality.

1.2 Representation alignment

But why is training diffusion-transformer architectures expensive in the first place? One answer comes from Yu et al. [5], who argue that these models suffer from a *representation bottleneck*. During training, a model is implicitly asked to learn two distinct tasks simultaneously: (1) representation: extracting semantically meaningful encodings from complex data; and (2) generation: constructing data from these encodings. The standard denoising objective alone may not provide sufficient signal for representation learning, leading to slow convergence and the large training budgets these models require in practice. In the context of molecular generation, this representation burden is compounded by the twin challenges outlined above. Models are now forced to encode complex structural geometry from datasets that only sparsely sample the true underlying distribution, all while operating without the aid of architectural priors.

Yu et al. propose Representation Alignment (REPA) as a remedy. The technique adds a regularisation term that aligns a generator's intermediate hidden states with the embeddings of a frozen pretrained encoder. On image diffusion, they report roughly a 17.5×

training speed-up against a strong baseline to achieve similar generation quality. REPA has since been extended beyond images to video [9], audio [10], and recently to molecular generation [7, 8], where it has been demonstrated but not systematically characterised.

1.3 Contributions

In this light, the core question we investigate in this report is: *can prior-light molecular generation models benefit from representation alignment with pre-trained models, in training efficiency and generation quality?* We aim to characterise when and why this works, including failure modes and limitations.

We document two lines of enquiry.

(i) *Does REPA work in the molecular domain?*: We conduct empirical studies on training flow-based generators with REPA in two regimes: Tabasco [15] for small molecules, and Proteina [14] for protein backbones. We find that REPA does not measurably improve Tabasco, but it improves training efficiency and advances the generation quality envelope for Proteina.

(ii) *When does REPA work?*: Following Singh et al. [6] in the realm of image generation, we develop a profiling framework to characterise the suitability of an encoder as a representation target. We measure three properties of an encoder: (1) an upper bound on the structural information it contains (linear-probe recoverability against domain structural labels); (2) a lower bound on what a projector can transfer from that information (projector-based recoverability); (3) the geometric conditioning of its embeddings for the alignment objective (effective rank).

We do not propose a new architecture or alignment loss, and our two studies are not a controlled experiment over the conditions that govern REPA's effectiveness. Instead, they are best read as paired case studies in regimes where those conditions differ.

The remainder of this report is organised as follows. Chapter 2 sets out the background and context needed to read the rest: flow matching, REPA, molecular generation. Chapter 3 discusses what it means for a generative model to be "good" in our domain, and the tensions that question hides. Chapter 4 develops the encoder-profiling framework and uses it to predict where REPA may and may not help. Chapters 5 and 6 present the two paired studies, small molecules first and protein backbones second. Finally, we conclude in Chapter 7 with a discussion of whether what we learn generalises beyond molecular generation.

Chapter 2

Background and Related Work

In this chapter, we introduce the machinery the rest of this thesis depends on. We outline flow matching (§2.1), the generative paradigm underlying the models we study, and representation alignment (§2.2), the regulariser that is our object of study. We then turn to the molecular setting. We situate generative models within the broader de novo molecular design pipeline (§2.3), before presenting the two molecular domains we work in, small molecules and protein backbones, and the flow-matching models we extend, Tabasco and Proteina (§2.4). The treatment is cursory by design. We intend only to guide the reader through this report, and we provide citations for omitted details.

2.1 Flow matching

At first glance, the term *generative modelling* implies the problem of *creating* new artefacts from scratch. However, it is perhaps more accurately described as the problem of *sampling* new artefacts from a complex data distribution that can only be accessed through a finite training set. A common strategy here is to fix a simple distribution we can sample from cheaply — typically standard Gaussian noise $p_0 = \mathcal{N}(0, I)$ — and learn a mapping from it to the desired data distribution p_1.

Flow matching (FM) [2] realises this mapping as a continuous-time transport. The model designer defines a *probability path* between a noise sample and a data sample — a continuous interpolation between them — and the model learns to carry the noise sample along this path. Concretely, it parameterises a time-dependent *velocity field* $v_\theta(x, t)$ that gives the instantaneous direction of motion at each location x and time $t \in [0, 1]$. At inference, integrating the learned ordinary differential equation (ODE) $\dot{x}_t = v_\theta(x_t, t)$ from $t = 0$ to $t = 1$ pushes a fresh noise sample forward into a fresh data sample.

A common choice of probability path is the linear interpolant, $x_t = (1 - t)\, x_0 + t\, x_1$ for $x_0 \sim p_0$ and $x_1 \sim p_1$. The model's target is the *marginal* velocity field at (x, t): the average direction of motion over all noise-data pairs whose paths pass through that point. However, supervising this marginal directly is intractable, since the contributing pairs are not known in advance. Flow matching sidesteps this difficulty with a key observation [2]:

although the marginal field may be intractable, the *conditional* velocity along the path of any specific noise-data pair is relatively trivial. Indeed, for the linear interpolant, it is simply $\dot{x}_t = x_1 - x_0$. Regressing the model on these conditional velocities, averaged over many sampled pairs, recovers the marginal field in expectation, and gives the training loss as a mean-squared regression:

$$\mathcal{L}_{\mathrm{FM}} \;=\; \mathbb{E}_{t,\,x_0,\,x_1} \left\| v_\theta(x_t, t) - (x_1 - x_0) \right\|^2 .$$

In practice, every training step samples a time $t \in [0, 1]$ and a noise-data pair (x_0, x_1), computes the interpolant x_t, and takes a gradient step on this squared error.

Flow matching with this linear interpolant generalises score-based diffusion: the variance-preserving Gaussian-path objective of denoising diffusion models [1] is a special case of the FM family [2, 4].

Generating samples from a trained flow-matching model amounts to integrating the learned velocity field. This is often performed with a standard ODE solver, with Euler steps usually sufficing for stable trajectories. The field also admits stochastic sampling variants — such as those drawing on the connection to score-based stochastic differential equations (SDEs) — which inject noise during the integration steps to explore the probability space and improve sample diversity. The models we work with in this report, Tabasco and Proteina, support both deterministic and stochastic samplers.

Conditional generation augments v_θ with context tokens — such as a class label, a sequence length, or a structural motif — with conditioning supplied at both training and sampling time. In this report, we focus on unconditional generation only.

For a more complete treatment of flow matching, we refer the reader to Lipman et al. [2] for the original presentation, and to the recent Lipman et al. guide [4].

2.2 Representation alignment

Chapter 1 introduced the argument made by Yu et al. [5] about the *representation bottleneck* in diffusion-transformer models. Their analysis applies to flow-matching models as well, given the equivalence between the two objectives discussed above (§2.1). The flow-matching objective implicitly asks a model to learn two things at once: a useful representation of the data, and a velocity field that interprets this representation. This twin burden, they argue, leads to slow convergence and large training budgets.

Their proposed solution, Representation Alignment (REPA), addresses the first half of this burden by adding a regularisation term to the training loss. Figure 2.1 shows the construction. The regularisation pulls z_ℓ, the generator's intermediate hidden state at a chosen *alignment layer* ℓ, towards the embeddings produced by a frozen pretrained encoder f_{enc} run on the clean data sample x_1. The hidden state z_ℓ is a sequence of N token vectors. In the context of images, each token represents an image patch; in the molecular

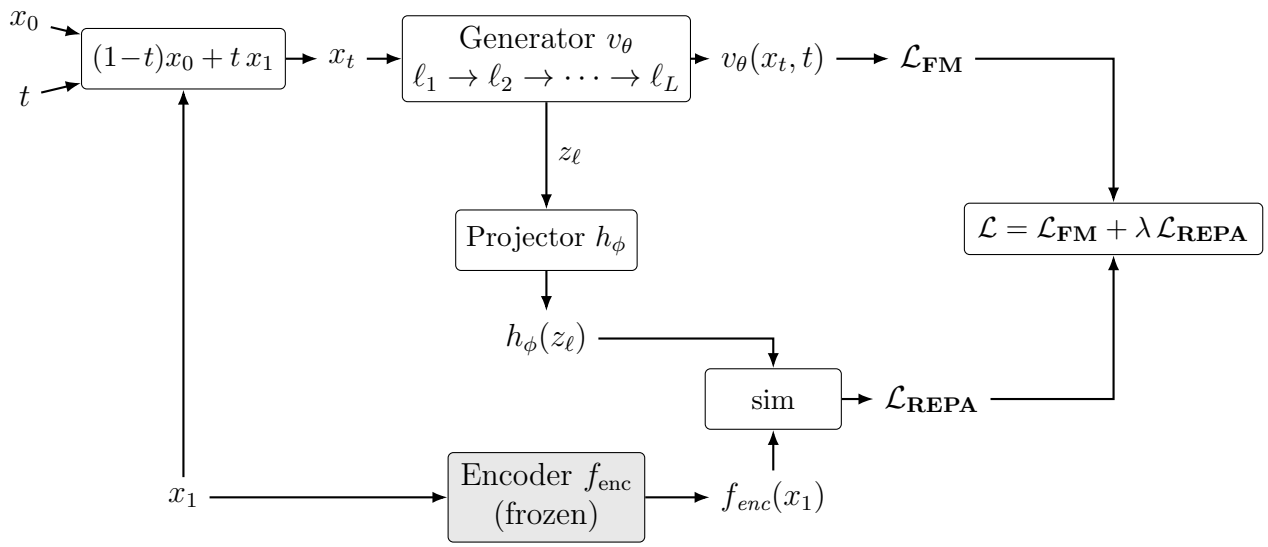

Figure 2.1: The REPA construction. A noised sample $x_t = (1-t)x_0 + t\,x_1$ is constructed at each training step from a noise sample x_0, a clean data sample x_1, and a sampled timestep $t \in [0, 1]$. The generator v_θ processes x_t through its sequence of layers $\ell_1, \ldots, \ell_L$ to produce the velocity field that feeds the flow-matching loss $\mathcal{L}_{\text{FM}}$. Separately, the clean sample x_1 is also passed through a frozen pretrained encoder f_{enc} (shaded). The generator's hidden state z_ℓ at the chosen alignment layer is mapped into the encoder's feature space by a trainable projector h_ϕ; the REPA loss $\mathcal{L}_{\text{REPA}}$ is the negated similarity between $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$, averaged across tokens. The total training loss $\mathcal{L}$ combines them with weight λ.

setting, each token represents an atom or residue that is part of the larger molecule (§2.3). A small trainable *projector head* h_ϕ — a three-layer multi-layer perceptron (MLP) in the original paper — maps z_ℓ into the encoder's feature space, so that $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$ can be compared. This alignment is asymmetric. The encoder provides a fixed target throughout training, while the generator parameters θ and the projector ϕ are updated.

The alignment loss is the negated similarity, averaged token-wise, between projected and target embeddings:

$$\mathcal{L}_{\text{REPA}} \;=\; -\,\mathbb{E}_{t,\,x_0,\,x_1}\left[\frac{1}{N}\sum_{n=1}^{N}\text{sim}\big(h_\phi(z_\ell)_n,\; f_{\text{enc}}(x_1)_n\big)\right],$$

where n indexes the N tokens within a sample. In the image setting, N is constant across the dataset, so this token-wise average is equivalent to averaging similarities once per sample and then across the batch. Indeed, the original paper draws no distinction between the two. For molecular data, where N varies per sample, the two diverge. We return to this distinction in the design knobs below.

The final training objective is the flow-matching loss from §2.1 plus this term, scaled by a weight λ:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{FM}} \;+\; \lambda\,\mathcal{L}_{\text{REPA}}.$$

This construction exposes several design choices, some of whose effects we revisit in the ablations of Chapter 6:

- The choice of encoder f_{enc}. Different encoders may capture distinct aspects of the data distribution.
- The alignment layer ℓ at which we attach the projector. Yu et al. experimentally determined that mid-network layers work best for image diffusion, but the optimal choice is not obvious a priori.
- The alignment-loss weight λ, which balances the alignment signal against the generative objective. Once again, Yu et al. experimentally determined an optimal value of 0.5 for image diffusion, but this may not hold universally.
- The choice of similarity metric sim. Yu et al. compared cosine similarity and mean-squared error for image diffusion, and found the former more effective. We adopt cosine similarity in this report as well.
- The averaging mode. Averaging tokens within each sample first weights every molecule equally, while pooling tokens across the batch gives larger molecules proportionally more REPA signal.

We adopt the core REPA construction as in Yu et al. for this report. However, a small family of methodological variants on REPA has since emerged. Dispersive regularisers [12] replace the external encoder with an internal contrastive term, while flexible-target formulations [7] broaden which encoder features the loss attaches to. Separately, REPA-style alignment has been applied beyond the image setting, to video [9], audio [10], materials [11], and molecules [8]. We defer exploration of these variants to future work.

2.2.1 What makes for a good REPA encoder?

There are many possible choices for the encoder f_{enc} in any REPA construction, and this selection is a key design decision. But what characteristics make for a good alignment target? In their original report, Yu et al. [5] observed that ImageNet linear-probe accuracy correlates positively with generation performance, suggesting that global semantic representation is the driving factor. However, more recent work by Singh et al. [6] reports the opposite after profiling a wider range of encoders. Linear-probe accuracy does not in fact predict REPA gain, they argue, while a measure of the spatial structure of patch-token representations does. This suggests that generation performance for REPA may be driven by *local* feature geometry, rather than *global* semantic strength. Encoder characterisation for REPA remains an open research question, particularly in the molecular generation setting, and we pick up the thread in Chapter 4.

2.3 A primer on molecular generation

Having outlined the modelling paradigms driving our report, we now turn to the molecular setting in which we apply them. To situate our work, we begin with a brief primer on the role generative models play within the broader de novo molecular design pipeline.

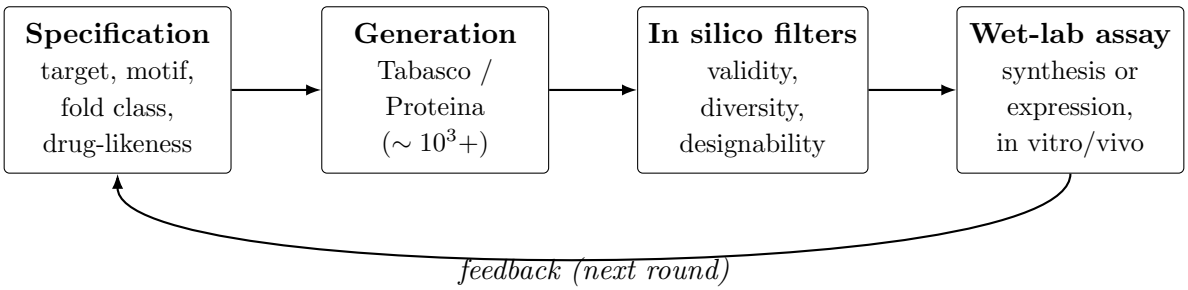

Figure 2.2: The de novo molecular design cycle. An upstream specification drives a generative model to produce a candidate pool, which is funnelled through in silico filters and then synthesised (or expressed) and assayed in the wet lab. Experimental results feed back into the next round of generation. Typical generation-step pool sizes are in the thousands to tens of thousands of candidates per round [18].

Generative models are rarely intended to produce ready-to-use artefacts. Instead, they act as high-throughput in silico proposal engines in an iterative process of refinement and selection. This process, often called a *design campaign* [53], is summarised in Figure 2.2.

An upstream specification, such as a drug-likeness profile or a fold class, conditions a generative model to produce a pool of candidate samples. These pools typically comprise thousands to tens of thousands of candidates per round [18], which are then funnelled through in silico filters. For small molecules, these typically include checks for structural validity and physical plausibility [54], drug-likeness [55], and target docking; for protein backbones, designability and structural diversity are the dominant considerations [14]. The select few survivors are produced in a laboratory, through chemical synthesis (small molecules) or expression in cell lines (proteins), and evaluated using in vitro or in vivo assays. Experimental results then feed back into the next round of generation.

This pipeline framing is relevant to our discussion of evaluation metrics in Chapter 3. Several metrics we adopt, such as structural validity and designability. are not measures of intrinsic molecular quality, but proxies for specific in silico filter steps.

2.4 Tabasco and Proteina

The two models we study — Tabasco [15] for small molecules and Proteina [14] for protein backbones — are both non-equivariant transformer flow-matching generators. They are explicitly designed to scale with data and model size, rather than relying on structural priors imposed by model architecture. The rotational and translational symmetries expected of 3D molecules are intended to be learned from data and augmentation rather than imposed by construction.

Tabasco is motivated by the success of similar architectures in protein-structure generation, like Proteina, setting it apart from equivariant message-passing small-molecule generators such as EQGAT-diff [48], MiDi [49], and SemlaFlow [50]. In turn, Proteina is motivated by the pair-bias attention mechanism introduced in AlphaFold 3 [44], which

distinguishes it from SE(3)-equivariant backbone generators such as RFdiffusion [18], FrameFlow [46], and Genie [47]. Despite their architectural minimalism, both achieve performance comparable or superior to equivariant baselines on structural-quality evaluations.

2.4.1 Small molecules and Tabasco

In medicinal chemistry, a *small molecule* is a drug-like compound at the sub-900-Dalton scale [51]. In silico, its structure is typically represented using two coupled features: a discrete *molecular graph* of typed atoms connected by typed bonds, and a continuous *conformer* placing those atoms in 3D Cartesian space. A flow-matching parameterisation must therefore handle both jointly.

Tabasco [15] represents each molecule as a sequence of per-atom tokens, each carrying an atom-type embedding and a 3D coordinate, but with no explicit bond decoder. This treats the molecule as a point cloud of typed atoms rather than a graph, sidestepping edge modelling at the cost of having to recover bond structure post hoc from interatomic distances. Tabasco is trained on GEOM-drugs [35], a public dataset of drug-like conformers.

This point-cloud-of-atoms parameterisation is one of several options in the literature, and each makes a different trade-off. *SMILES* and *SELFIES* are string encodings that work well with autoregressive models but discard the 3D conformer entirely; we encounter them later as the parameterisation employed by some candidate encoders we evaluate in Chapter 4. *3D voxel grids* preserve geometry, but lose the discrete graph and inflate the token count by an order of magnitude or more. *Equivariant point clouds*, used by generators such as EQGAT-diff [48], MiDi [49], and SemlaFlow [50], keep both graph and geometry, but pay the computational cost associated with SE(3)-equivariant layers. We refer the reader to Duval et al. [39] and Wang et al. [22] for broader surveys in this space.

2.4.2 Protein backbones and Proteina

A *protein* is a chain of amino-acid residues, ranging from around 50 to several thousands in length [52]; the largest known, titin, has approximately 34,000 residues. (Shorter chains are typically classified as *peptides* instead.) Each amino-acid residue contributes four *backbone* atoms (N, C_α, C, O) whose interactions determine the chain's *fold* geometry, and a *side chain* that determines its amino-acid identity.

A protein's *secondary structure* is its local pattern of α-helices and β-strands; its *tertiary structure* is the global three-dimensional fold those elements form when they pack together. A CATH code [58] (Class, Architecture, Topology, Homology) classifies this fold hierarchically. *Class* captures broad secondary-structure content (mostly-α, mostly-β, or mixed), *Architecture* the gross spatial arrangement of those elements, and *Topology* their specific connectivity.

For the structure-generation problem we study, it is customary to only model the back-

bone, with sequence identity recovered downstream using a separate *inverse-folding* model such as ProteinMPNN [24]. A further compression is the *CA-trace*, which uses only the set of C_α coordinates for every residue to represent the entire protein. This parameterisation retains enough information to recover fold topology and reduces computational cost relative to the more complete representation, which is typically referred to as *all-atom* modelling. Proteina [14] adopts this C_α parameterisation, representing each backbone as a sequence of per-residue tokens carrying continuous C_α Cartesian coordinates. The models reported in the original presentation are largely trained on high-confidence synthetic structures from the AlphaFold Database (AFDB) [34].

Once again, the CA-trace parameterisation is one of several techniques used in the literature to parameterise proteins in silico. *Sequence-only* models, such as the ESM lineage [23], operate on amino-acid identity strings. These models capture rich functional priors, but encode far less information on 3D structure. *Full-atom* representations close the gap to physical realism, but add substantial compute cost. *Equivariant* parameterisations, such as torsion angles and oriented residue frames, encode SE(3) symmetries by construction, and form the standard basis for the equivariant generators noted above. However, these architectures have historically bottlenecked scaling. We refer the reader to Notin et al. [56] for a broader survey of machine-learning approaches to protein design.

2.5 Open questions

This survey leaves two questions open that the rest of this report takes on:

(i) *Does REPA work in the molecular domain?* Existing applications of REPA in molecular generation [7, 8] have not been ablated over encoder choice, and protein backbone generation in particular remains unexplored.

(ii) *When does REPA work?* The encoder-characterisation work surveyed in §2.2.1 has begun to ask which properties of an encoder predict REPA gain, but only within the image setting. We extend this question to the molecular domain.

We take both questions up empirically in Chapters 4–6.

Chapter 3

Evaluating molecular generation

Chapter 4

Encoder targets and profiling

Chapter 5

REPA on small molecules

Chapter 6

REPA on protein backbones

In this chapter, we expand the scope of our study on representation alignment from small molecules to much larger ones: proteins. Previously (Chapter 5), our efforts using REPA to generate small molecules yielded little demonstrable improvement, largely because the Tabasco [15] baseline was already saturated on most measures of generation quality. However, de novo backbone design represents a more challenging task, as the data distribution of proteins is significantly larger, and the geometry of these structures is correspondingly more complex. In this context, we ask: *can REPA accelerate or improve the training of Proteina [14], a flow-matching transformer-based protein backbone generator?*

Our headline finding is that REPA accelerates convergence on the two key measures of protein generation quality. This holds across both experimental and synthetic training-data regimes (Figure 6.1). This chapter characterises *what* happens and attempts to understand *why*. We also pick up the thread from Chapter 3 on how the molecular domain differs from the image domain REPA was built for, and develop a theme throughout: REPA here is a *family* of interventions rather than a single mechanism.

6.1 Experimental setup

We integrate REPA into Proteina [14], a 60M-parameter non-equivariant flow-matching generator of Cα protein backbones. Following §2.2, we attach a trainable projector at a chosen layer of the generator trunk (the transformer stack) that aligns the model's hidden states to a frozen external encoder.

6.1.1 Datasets

We train in two regimes that span the field's experimental–synthetic spectrum: the experimentally-determined Protein Data Bank [33], and the AlphaFold-predicted AlphaFold Database [34] (Swiss-Prot subset). (We use the manually-curated Swiss-Prot subset of AFDB as a reproducible stand-in for the Foldseek-clustered TrEMBL subset on which Proteina was trained in the original presentation [14]. This corpus contained data that has since been pruned from AFDB, and is hence inaccessible.)

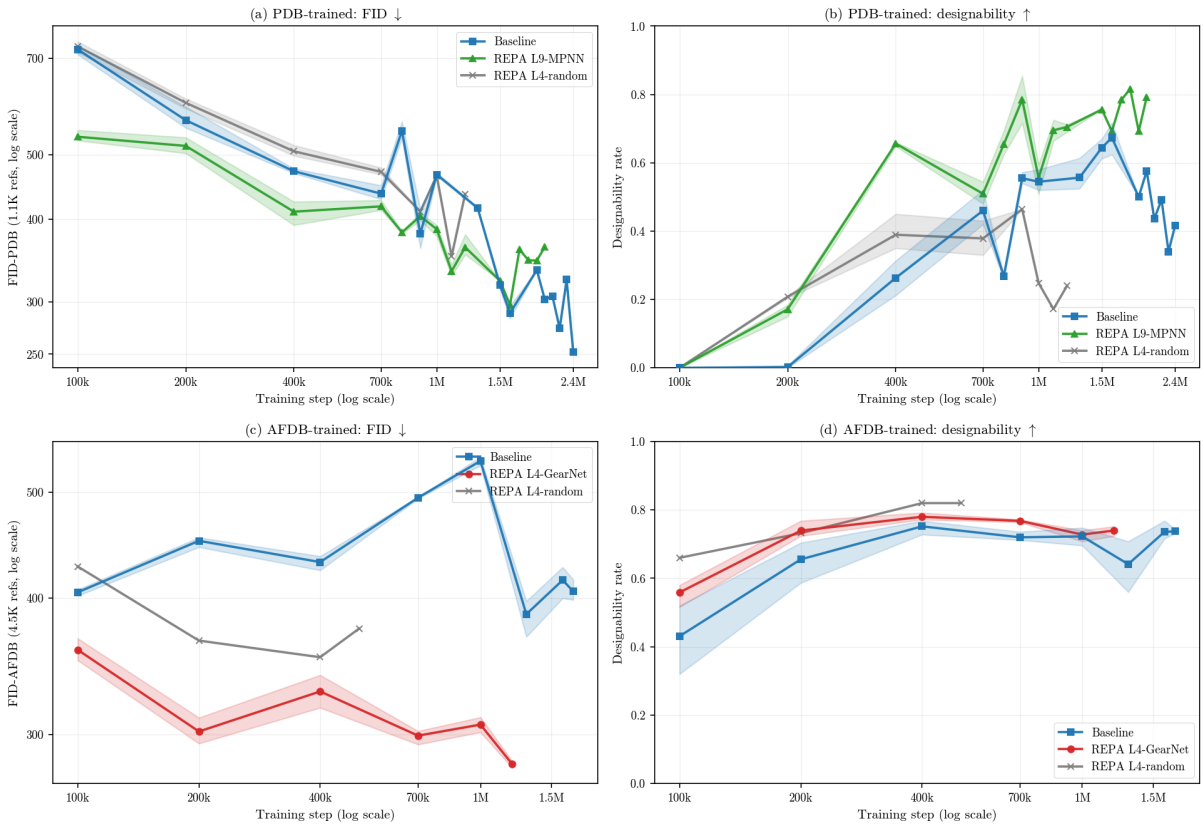

Figure 6.1: **REPA accelerates both FID and designability, in both PDB (experimental) and AFDB (synthetic) data regimes.** The random-encoder control (grey) helps distinguish gains induced by regularisation alone from those induced mechanistically by an encoder's learned structure. While PDB retains a persistent designability gap, its baseline appears to catch up on FID after ≈1M steps; we examine this in §6.2.3. ($n \leq 256$ models, 3 seeds; shaded bands show min/max.)

Following Proteina [14], we train on the subset of chains with at most 256 residues; we represent this length-regime by $n \leq 256$. (We note that this represents only a property of the training datasets, not a bound on the length of the proteins that models trained on these datasets can generate.) Following this filter, the two training sets are comparable in size (PDB: 273,771 / 3,190 / 323 train/val/test chains; AFDB: 229,670 / 4,521 / 235). Appendix A (Table A.1) contains further details on dataset composition.

Different datasets in this domain carry different biases. Trends in designability offer a clear example, as training on AFDB raises the in-silico-designability floor well above PDB at every setting we tested (§6.2.3). The likely cause is circular. The canonical designability test inverse-folds a generated backbone with ProteinMPNN [24] and re-folds it with ESMFold [23] (Chapter 3). Both of these models share their lineage with the AlphaFold2 [43] network that produced AFDB in the first place, so AlphaFold-like backbones are plausibly the ones it re-folds most confidently. We do not test this mechanism, but flag the observation so metrics are read with care.

Property	Value
Base architecture	
Backbone (*trunk*)	Proteina (CAFlow): 60M-param, 10-layer non-equivariant Cα transformer (token dim 512, 8 heads)
REPA alignment	
Encoder targets	GearNet (512-d) · ProteinMPNN (128-d) · random-GearNet (control)
Injection depth ℓ	layer 4 (middle) · layer 9 (last)
Projector	3-layer MLP, hidden dim 512
Similarity metric sim	cosine similarity
Averaging mode	per residue
Alignment weight λ	0.5
Training & sampling	
Learning rate	10^{-3}
Batch size	24 ($n \leq 256$)
Weight decay	none
Sampler	SDE, noise scale $\gamma = 0.45$

Table 6.1: Model and training configuration. This chapter presents two encoder targets and two injection depths. Other ablations are deferred to Appendix A.

6.1.2 Models

The base Proteina model comprises a stack of ten transformer layers (zero-indexed) that feeds a flow matching objective. Our experiments add a REPA loss to this design, as outlined in Table 6.1.

Encoder and injection-depth ablations. Our narrative concentrates on the two encoders identified as most viable in Chapter 4: *GearNet* [25], a structure encoder pretrained for CATH fold classification (checkpoint released with Proteina [14]); and *ProteinMPNN* [24], which encodes each residue's local inverse-folding environment. Each encoder is attached to the trunk in two layer-depth configurations: layer 4 (middle layer) and layer 9 (last layer). The remaining encoder and depth combinations are collected in Appendix A. Figures presented in this chapter will typically follow the best performing model to serve as an illustrative example.

Separating regularisation from mechanism. A random-weights GearNet attached at layer 4, identical in architecture but never trained, serves as a control architecture in this chapter. We use this model to separate whatever REPA achieves merely due to the regularisation effect of adding an auxiliary loss from what it achieves through the encoder's *learned* structural knowledge. Crucially, a trained encoder at the same layer 4 still outperforms the random one, so the learned-versus-random gap is not an artefact of injection depth (Appendix A).

Hyperparameters. Our defaults largely follow the original image-domain REPA recipe [5].

6.1.3 Metrics

The original hypothesis that motivated REPA for imaging [5] was that alleviating a *representation* bottleneck would improve *generation* quality. We seek to understand if the same mechanism applies in the molecular domain as well. Hence, along with estimating the quality of a model's generated samples, we are also interested in what the model comes to *represent* internally. To this end, we measure three families of metrics, as outlined below: representation quality, representation alignment, and generation quality (Table 6.2).

Representation quality asks what domain-specific information is *linearly decodable* from the trunk's hidden states, following the probe principle of §3. We adapt three probing tasks from the literature, spanning three levels of abstraction:

1. *per-residue inverse-folding (IF) sequence recovery* [27], to understand whether the representation recovers the local chemical environment;
2. *per-residue backbone dihedral angle error* [26], to understand whether it encodes low-level geometry; and
3. *per-chain CATH fold classification* [26], to understand whether mean-pooled residue representations carry information about global fold structure (the CATH hierarchy is defined in §2.4.2). We headline on architecture (CATH-A), as Class offers arguably too easy a discrimination task given top-1 saturation, and Topology offers too many buckets and too few data points for a meaningful evaluation. However, we report all three levels for completeness, noting that the patterns we observe hold true across them, and motivate our choices in Appendix A (Table A.2).

We note that these three tasks probe largely independent capabilities. Local sequence, local geometry, and global fold are orthogonal to one another, so even a model that is optimised for one may not perform well on the others. Hence, improving performance on all three at once is strong evidence of holistic representation quality.

Representation alignment asks how close one model's representations are to those produced by another — here, the trunk versus a structural encoder. We measure it with CKNNA [59], a mutual-nearest-neighbour kernel alignment between two representations, as used in the original REPA analysis [5] and a recent molecular extension [8]. We use alignment as a diagnostic — a mechanistic lens on *how* REPA might produce its representation gains — rather than a quantity we optimise or build claims on.

Generation quality is what ultimately matters, and we report it in the five groups of Chapter 3: per-sample quality, tertiary structure over the whole set and the designable subset (T-W, T-D), and secondary structure over the same two (S-W, S-D).

A second axis cuts across these groups and recurs through the chapter. *Fidelity* metrics ask how closely the generated distribution matches a reference: FID, the fold Jensen–Shannon divergence (fJSD), and our secondary-structure divergence (ssJSD-2D). *Diversity* metrics ask how widely the generated set spreads: the fold score (fS), pairwise TM-score,

and cluster count. Most of these follow the original presentation in Proteina [14], which introduced distributional (FID-style) metrics to the protein-generation literature.

Building on these, we introduce a new metric, ssJSD-2D, which compares the full (helix-fraction, strand-fraction) secondary-structure distribution between a set of proteins and a reference dataset. A fuller description is given in Appendix A.4. As we will explore in §6.2.4, REPA improves fidelity and whole-set diversity together, but may trade off tertiary-structure diversity *within* the designable subset.

6.1.4 Evaluation protocol

Representation probing. To measure the representation quality of a frozen model checkpoint, we fit linear probes to the hidden states it generates for a batch of clean inputs (t=1). The inverse-folding and dihedral probes are per-residue (a linear classifier and a linear regressor); the CATH probe is a per-chain logistic regression on mean-pooled representations. The classification probes (inverse folding and CATH) are scored as top-1 accuracy. IF is a 20-way problem, while CATH-A is a ∼40-way problem. The dihedral probe reports mean absolute angular error.

Each probe is fit on 5,000 training chains. We evaluate the per-residue probes (inverse folding and dihedral) on a *blinded* set of ∼325 proteins (∼43k residues) held to under 30% sequence identity to all training data, which removes both model- and probe-side leakage. The per-chain CATH probe needs many distinct chains spread across fold classes, which that small slice cannot supply, so we relax it to a *probe-clean* evaluation on a larger validation set of ∼3,190 chains. We detail this scheme, and the model/probe leakage distinction it controls for, in Appendix A.3, and read rank-order and Δ-from-baseline throughout this chapter.

Representation alignment. We measure how close a model's geometry is to an encoder's using the CKNNA [59] metric, using clean (t=1) data from a fixed held-out batch. We compute this metric between the hidden states at every layer in the trunk and three domain-relevant encoders: GearNet, ProteinMPNN, and ESM2 [23].

CKNNA compares two representations using their neighbourhood structure. For each item, it takes the k=10 nearest neighbours in each representation, keeps only the pairs that are neighbours in both, and scores how well the two agree on them. Hence, it measures local structure rather than global geometry. The score is normalised so that self-alignment is 1 and alignment to random features is ≈ 0.

We compute CKNNA per-residue using 10,000 residues. Each value is bracketed by a subsample-without-replacement bootstrap: 50 draws at 80% of the sample. (Sampling with replacement would inflate the metric through duplicate nearest-neighbour pairs.) We note that the absolute CKNNA values we record are small, an order of magnitude below image-domain REPA [5]. We therefore read the direction and layer structure of the effect, and its separation from the baseline, not its raw magnitude.

Metric	Definition	Content	Granularity
Representation quality			
IF top-1 ↑	Inverse-folding sequence recovery (top-1 accuracy)	Semantic content	Per residue
Dihedral MAE ↓	Backbone dihedral-angle error	Low-level geometry	Per residue
CATH-A top-1 ↑	Fold-architecture classification top-1 accuracy (Table A.2)	Fold-level structure	Per chain
Representation alignment			
CKNNA ↑ [59]	Mutual-nearest-neighbour kernel alignment	Trunk–encoder similarity	Per residue
Generation quality (Chapter 3)			
Quality			
FID ↓	Frchet distance to the reference set (GearNet features)	Distributional fidelity	Whole set
Des ↑	Fraction of samples that are designable	Physical realisability	Whole set
Tertiary structure — whole set (T-W)			
fJSD-A ↓	Jensen–Shannon divergence vs. reference CATH-A distribution	Fold-distribution fidelity	Whole set
fS-A ↑	Spread of generated folds across CATH-architecture classes	Whole-set fold diversity	Whole set
pwTM ↓	Mean pairwise TM-score	Structural diversity	Whole set
Tertiary structure — designable subset (T-D)			
#Clust ↑	Number of TM ≥ 0.5 structural clusters	Designable diversity	Designable subset
pwTM ↓	Mean pairwise TM-score, designable subset	Designable diversity	Designable subset
Secondary structure — whole set (S-W)			
ssJSD-2D ↓	Jensen–Shannon divergence of the (helix-frac, strand-frac) distribution	Secondary-structure fidelity	Whole set
Secondary structure — designable subset (S-D)			
ssJSD-2D ↓	As above, on the designable subset	SS fidelity (designable)	Designable subset
E%des ↑	Mean β-strand (E) fraction	β-content	Designable subset

Table 6.2: The three measurement families used in this chapter: representation quality, representation alignment, and generation quality (five reported groups). The whole set is all generated backbones; the designable subset is those passing the designability filter. Arrows give the good direction. Full per-metric results appear in Tables 6.6 and 6.7.

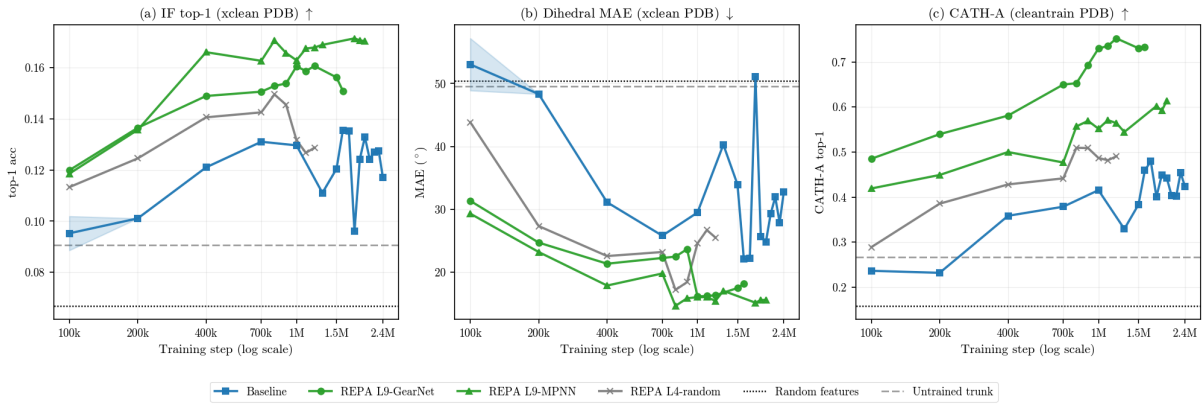

Figure 6.2: **REPA offers a mechanistic effect that lifts the trunk's representation quality asymptotically.** REPA variants open a quality gap over baseline that never closes. The random control gains least, evidence that a trained encoder contributes a genuine mechanistic effect rather than mere regularisation from the auxiliary loss. ($n \leq 256$, PDB-trained; best-layer linear probes; per-residue probes on a blinded cross-database slice, CATH on a probe-clean split. See Appendix A.3 for further details and ablations.)

Generation sampling. Our generation evaluation follows the Proteina protocol [14], at a reduced sample budget.

For whole-set evaluations, we sample 1,125 backbones per checkpoint — 125 at each length in $\{50, 75, \ldots, 250\}$ — and compute metrics over all of them. Depending on the training regime, we use matching PDB or AFDB reference statistics.

For designability evaluations, 50 backbones at each length $\{50, 100, 150, 200, 250\}$ are inverse-folded with ProteinMPNN (8 sequences each), re-folded with ESMFold, and called designable when the self-consistency RMSD is below 2 Å. The designable-subset metrics (T-D, S-D) are then computed on the backbones that pass.

With this experimental testbed in place, the rest of the chapter investigates what the REPA construction changes with respect to baseline, and how that helps improve generation quality. We now turn to the results.

6.2 Results

6.2.1 REPA improves representation quality asymptotically

At baseline, the flow-matching objective helps the model encode some useful representational information. We probe our models along the three orthogonal axes of §6.1.3 — local chemical environment (inverse folding), local geometry (backbone dihedrals), and global fold (CATH classification, with CATH-A as our headline metric; Figure 6.2). The trained baseline consistently does better than both random features and an untrained Proteina trunk. This suggests that the flow-matching objective does indeed learn to *represent* protein backbones as it learns to *generate* them.

REPA improves representational capacity, consistently and permanently. Ev-

Variant	IF top-1 ↑	dihedral MAE (°) ↓	CATH top-1 acc. ↑		
			C	A	T
Baseline	0.130	27.6	0.717	0.397	0.303
REPA L4-random	+0.007	−5.1	+0.101	+0.089	+0.060
REPA L9-GN	+0.026	−8.1	+0.171	+0.305	+0.449
REPA L9-MPNN	+0.036	−11.4	+0.120	+0.151	+0.172

Table 6.3: **REPA offers persistent and encoder-routed representation advantages.** Consistent with what each encoder was trained for, REPA-GearNet wins the fold probes (CATH C/A/T), while REPA-MPNN wins the per-residue probes (IF, dihedral). We headline on CATH-A, and report C and T for completeness. ($n \leq 256$ PDB-trained; best-layer linear-probe performance, averaged over the 700K–1.2M training-step window; Δ from baseline; green / darker = improvement/best per probe.)

ery REPA variant lifts all three probes substantially above the baseline from the very beginning of training, and the difference persists. This is an asymptotic gap, not just a head start or an accelerated convergence pattern. This gap confirms one of the hypotheses posited in the original REPA presentation [5]: an alignment target during training can fill the representational headroom left by the flow-matching objective.

Every encoder routes representation gain differently. REPA-GearNet consistently wins the fold probes, while REPA-MPNN wins the per-residue probes (Table 6.3). This pattern follows what we would expect a priori, given the tasks that each encoder has been specifically trained for. This ties into our emerging theme on the multi-factorial nature of representation learning in the molecular domain (Chapter 3). No single encoder captures every factor, so the choice of alignment target is also a choice over the representational factor a model designer wishes to emphasise. We establish this routing on PDB-trained models and report the AFDB results in Appendix A.

REPA offers a mechanistic effect beyond just regularisation. Our random-weights control also improves on the baseline across all probes. This may be because *any* regularisation generically improves the model's representation capacity, or because even an untrained message-passing network imposes geometric priors that aid the trunk. Nevertheless, trained encoders consistently offer more, suggesting a genuine mechanistic effect.

6.2.2 A diagnostic check on alignment

Following Yu et al. [5] and recent in-domain work [8], we investigate the mechanism that may be driving REPA's representation quality gains by quantifying the representation alignment it induces. Using CKNNA [59], we score how closely our models' per-residue geometry matches that of three domain-specific encoders: GearNet and ProteinMPNN, our two alignment targets in this chapter; and ESM2 [23], which we align to only in our ablations, not in the models presented here. We note that we study alignment only as a diagnostic, and not to make claims about REPA's utility.

REPA lifts alignment above the random floor, albeit at low magnitude. The

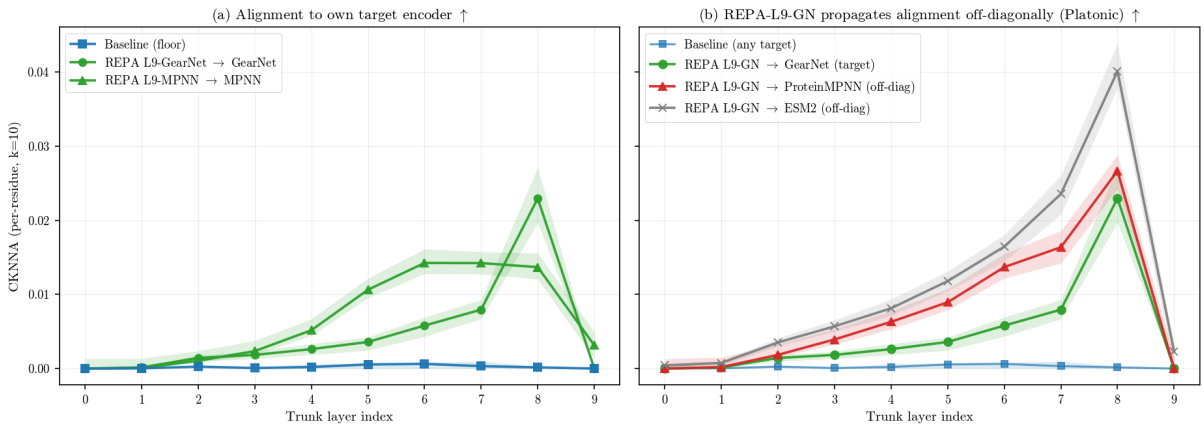

Figure 6.3: **REPA aligns the trunk with domain encoders, including those it was not trained with.** (a) Each REPA variant lifts per-residue alignment to its own target encoder above the baseline floor. (b) REPA induces cross-encoder alignment: for example, a GearNet-aligned trunk also moves toward ProteinMPNN and ESM2. Absolute alignment is small throughout, so we read the pattern, not its magnitude. ($n \leq 256$ PDB-trained, step 1M, t=1.0; per-residue CKNNA, k=10; lines are bootstrap medians, shaded bands are the 5–95% subsample bootstrap interval.)

baseline sits at the random floor at every layer (CKNNA ≈ 0), while REPA separates from it cleanly, suggesting that its hidden states do move towards the encoder's (Figure 6.3a). This serves as a minimal check that the construction changes model behaviour. However, the magnitude of this signal is small. Peak per-residue CKNNA (bootstrap median) is $\approx$ 0.02–0.05, on a scale where self-alignment is 1. We therefore read the pattern, not its size, and stake no strong claim on these numbers.

Aligning to one encoder nudges the trunk towards others as well. Every REPA variant we tested raises alignment to all three encoders, including those it was not aligned to (full model×encoder matrix in Appendix A). A GearNet-aligned trunk, for example, also moves closer to ProteinMPNN and ESM2 [23] (Figure 6.3b), suggesting that REPA-induced alignment is not merely circular self-alignment.

One possible explanation for this phenomenon is the *Platonic representation hypothesis* [59]. This idea suggests models trained on related data converge toward a shared representation. In this framework, REPA may be nudging the trunk toward a common structural representation that several encoders approximate, rather than toward any single encoder. We offer this only as a possible interpretation, not as a claim.

6.2.3 REPA accelerates generation quality by climbing the representation–generation envelope faster

In this subsection, we focus on the holistic *Quality* metrics of our suite, FID and designability (outlined in Table 6.2), to outline our headline results. A deeper analysis of generation quality along tertiary- and secondary-structure decomposition follows in §6.2.4.

REPA gives generation quality an early head start in every regime. At 400K

Variant (last ckpt)	**Acceleration** (@400K)	**Long run** (best)
AFDB FID ↓ — baseline (to 1.7M) @400K 431, best 386 within 2.0M		
GearNet-L4 (1.2M)	+24% (328)	+27% (282 @1.2M)
GearNet-L9 (900K)	+28% (313)	+19% (313 @400K)
MPNN-L4 (1.1M)	−10% (477)	−8% (416 @1.0M)
MPNN-L9 (1.5M)	+18% (352)	+9% (352 @400K)
Random control (500K)	+18% (353)	+9% (353 @400K)
AFDB designability ↑ — baseline (to 1.7M) @400K 0.75, best 0.75 within 2.0M		
GearNet-L4 (1.2M)	+4% (0.78)	+4% (0.78 @400K)
GearNet-L9 (900K)	−4% (0.72)	+3% (0.77 @800K)
MPNN-L4 (1.1M)	−9% (0.68)	−8% (0.69 @700K)
MPNN-L9 (1.5M)	−6% (0.71)	+2% (0.77 @700K)
Random control (500K)	+9% (0.82)	+9% (0.82 @400K)
PDB FID ↓ — baseline (to 2.4M) @400K 472, best 288 within 2.0M		
GearNet-L4 (1.0M)	+7% (440)	−50% (432 @1.0M)
GearNet-L9 (1.6M)	+28% (341)	−11% (319 @700K)
MPNN-L4 (1.6M)	−8% (512)	−14% (329 @1.6M)
MPNN-L9 (2.0M)	+13% (410)	−3% (297 @1.6M)
Random control (1.2M)	−7% (506)	−22% (351 @1.1M)
PDB designability ↑ — baseline (to 2.4M) @400K 0.26, best 0.67 within 2.0M		
GearNet-L4 (1.0M)	+98% (0.52)	+4% (0.70 @700K)
GearNet-L9 (1.6M)	+97% (0.52)	−7% (0.63 @800K)
MPNN-L4 (1.6M)	+124% (0.59)	+5% (0.71 @1.6M)
MPNN-L9 (2.0M)	+149% (0.66)	+21% (0.82 @1.8M)
Random control (1.2M)	+48% (0.39)	−31% (0.46 @900K)

Table 6.4: **REPA accelerates generation quality over the baseline.** In some regimes it also wins the long run. (Values are percentage deltas from the baseline, with the absolute score in parentheses; acceleration is measured at 400K, the anchor of Figure 6.4, and the long run as each variant's best within a common 2.0M-step window; generation scores are means over up to three sampling seeds; $n \leq 256$; green / red = improvement/regression.)

steps into training, most REPA variants lead the baseline on both metrics: FID by up to +28%, and designability by up to +149% (Table 6.4).

This acceleration is consistent with the representation-bottleneck hypothesis. Yu et al. [5] hypothesise that a model's representation quality and its generation quality are coupled, because it is learning both at once. In our models, this coupling is clear. Across checkpoints, representation and generation quality are positively correlated, and the correlation survives controlling for training step (Table 6.5). If the two move together, then securing good representations early should secure good generation early as well.

REPA climbs the representation–generation envelope faster than the baseline. REPA reaches an asymptotically higher representation quality almost as soon as training begins (§6.2.1, Figure 6.2). On the representation–generation curve posited by the bottleneck hypothesis, this places it further along than the baseline at matched compute (Figure 6.4). Hence, it reaches a given level of generation quality at less training. We interpret REPA as simply traversing this curve faster, rather than finding a new shortcut.

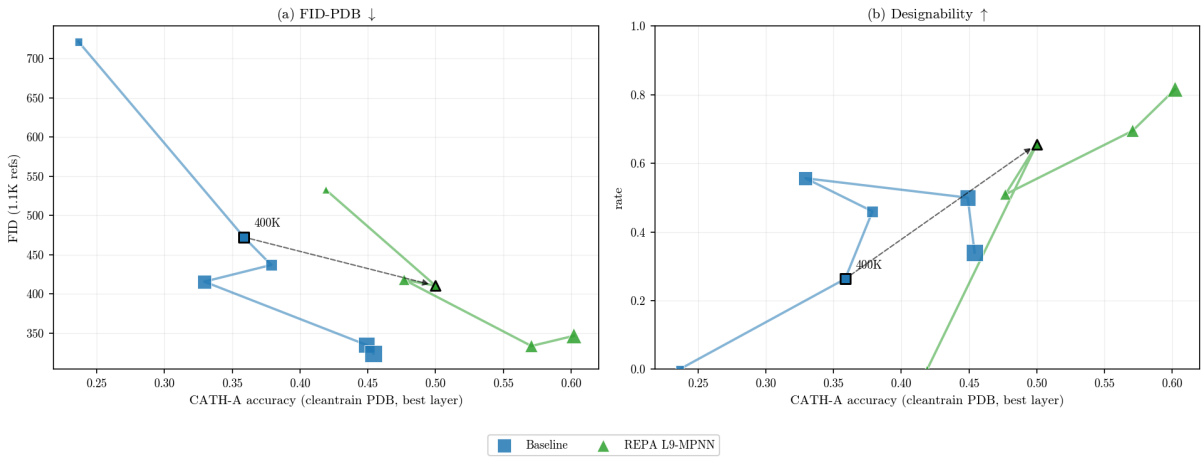

Figure 6.4: **At equal compute, REPA achieves better representation and generation quality.** The dashed arrow marks the matched-compute comparison at step 400K. Against the baseline, the REPA variant has higher fold accuracy (CATH-A) and better generation lower FID (a) and higher designability (b). (Markers are a coarse ≈400K-spaced subset of checkpoints from 100K, with the 400K point bordered; x = student CATH-A accuracy, higher to the right; generation values are means of N=3 sampling seeds; $n \leq 256$, PDB-trained.)

Representation quality	**FID**		**Designability**	
	partial	raw	partial	raw
Inverse folding (top-1)	+0.36	+0.34	+0.76	+0.70
Dihedral MAE	+0.37	+0.35	+0.60	+0.57
CATH-A	+0.22	+0.27	+0.48	+0.49

Table 6.5: **Better representations track better generation.** The effect is especially strong for designability, and moderate for FID. Partial Pearson correlation controlling for training step, with the raw value alongside; oriented so a positive value means better representation accompanies better generation. (N=59 checkpoints pooled over the baseline, the REPA variants, and the random control; $n \leq 256$, PDB-trained.)

This representation–generation link is encoder-matched, suggesting a causal relationship. Each encoder's generation gain lands on the axis where its representations are strongest. GearNet, the better fold encoder (§6.2.1), gives the larger gain in fold *distribution*; MPNN, the better local encoder, gives the larger gain in *designability*. A generic regularisation effect would be unlikely to produce this pattern, but we do not claim a formal proof of causality.

Broadly, REPA's lasting advantage tracks the headroom the baseline leaves. Where the baseline plateaus poorly, REPA's lead is large and permanent, as on AFDB FID. Where the baseline is a slow learner, the lead narrows yet still holds, as on PDB designability. Only where the baseline is a genuinely strong learner, as on PDB FID, does it eventually close the gap. AFDB designability is the exception, as it leaves little headroom over the baseline. However, as we noted earlier (§6.1.1), this is likely a quirk of the relationship between how designability is measured and how AFDB was generated,

Variant	Quality Des↑	Quality FID↓	T-W fJSD-A↓	T-W fS-A↑	T-W pwTM↓	T-D #Clust↑	T-D pwTM↓	S-W ssJSD2D↓	S-D ssJSD2D↓	S-D E%des↑
PDB-trained, n≤256, 700K, 3 seeds; FID is FID-PDB (in-distribution).										
Baseline	0.46	437	1.11	5.48	0.22	77	0.27	0.35	0.40	0.22
REPA L4-random	−0.08	+34	−0.55	+0.05	−0.05	−23	−0.10	−0.13	−0.01	−0.11
REPA L4-GN	+0.24	+71	−0.38	+1.63	+0.13	+1	+0.11	−0.21	−0.12	+0.00
REPA L9-GN	+0.14	−118	−0.57	+2.31	−0.03	+27	−0.02	−0.19	−0.10	−0.04
REPA L4-MPNN	+0.18	−88	−0.13	+0.48	+0.05	−4	+0.08	−0.16	−0.14	+0.01
REPA L9-MPNN	+0.05	−19	−0.32	+0.37	−0.01	+33	−0.01	−0.14	−0.11	−0.07
AFDB-trained, n≤256, 700K, 3 seeds (MPNN-L4: 1 seed); FID is FID-AFDB.										
Baseline	0.72	494	2.45	4.59	0.20	128	0.22	0.15	0.16	0.15
REPA L4-GN	+0.05	−195	−1.56	+2.27	+0.00	−35	−0.04	−0.11	−0.07	−0.02
REPA L9-GN	+0.01	−172	−1.29	+2.76	+0.01	−50	+0.11	−0.12	−0.06	+0.04
REPA L4-MPNN	−0.03	−22	−0.39	+0.03	—	−1	−0.07	−0.02	+0.05	−0.01
REPA L9-MPNN	+0.05	−139	−0.93	+0.31	−0.03	+21	−0.03	+0.01	+0.02	−0.04

Table 6.6: **Across the metric suite, almost every REPA variant improves whole-set fidelity and diversity over baseline.** The effect is broad enough that even the random-weights control matches the reference distribution better; only the learned encoders also lift per-sample quality (FID and designability). ($n \leq 256$, step 700K; baseline absolute, REPA rows show Δ; green / red = improvement/regression; header arrows mark the good direction.)

and not due to any special merit. Tellingly, even the random control scores highly here.

6.2.4 REPA improves whole-set coverage but may trade off diversity in the designable subspace

In this subsection, we investigate how REPA impacts different aspects of generation quality beyond our headline metrics. We read these observations as interesting qualifiers that provide further context to the main convergence result discussed above (§6.2.3).

REPA consistently improves whole-set fidelity and diversity metrics. Under almost every REPA variant tested, the model's learned data distribution more closely replicates the reference distribution on tertiary- and secondary-structure. Curiously, this is true of the random-weights control as well. However, only the learned encoders translate this advantage into higher FID and designability (Table 6.6).

However, a REPA-learned distribution may collapse onto fewer folds in its designable subspace. Across configurations, the number of distinct designable folds generated is often lower than baseline. This appears especially true for β-sheet-rich samples, where the drop-off is severe (Table 6.7). Crucially, the pattern holds even at matched β-content. Within the $\beta \geq 25$ bin, the baseline stays diverse (pwTM≈0.13) while REPA variants collapse onto near-identical folds (≈0.87). Somehow, the REPA construction appears to generate the *same* β-rich folds repeatedly, while the baseline spans many.

Fold concentration occurs early in training, and the baseline grows out of it. After 700K training steps, baseline and REPA models alike concentrate their β-rich designable samples. However, by 1.0M the baseline has diversified while REPA stays concentrated (Table 6.7). This suggests that REPA does not *engender* the concentration,

Designable-subset pwTM↓	700K steps			1.0M steps		
	β<10	10–25	β≥25	β<10	10–25	β≥25
PDB-trained						
Baseline	0.15	0.22	0.50	0.15	0.23	0.13
REPA L9-GN	0.16	0.27	0.73	0.14	0.24	0.87
REPA L9-MPNN	0.16	0.21	0.67	0.26	0.22	0.67
AFDB-trained						
Baseline	0.14	0.29	0.17	0.16	0.26	0.15
REPA L9-GN	0.37	0.31	0.82	0.40*	0.32*	0.76*
REPA L9-MPNN *(falsifier)*	0.15	0.27	0.11	0.15	0.24	0.15

Table 6.7: **A REPA-learned distribution may collapse onto fewer distinct folds within its designable subset.** The effect is strongest in the β-sheet-rich folds. Within the matched-β β≥25 bin, REPA concentrates onto near-identical folds, while the baseline stays diverse. Meanwhile, AFDB-MPNN-L9, which routes away from β-rich folds (E%des, Table 6.6), escapes fold concentration. (Designable-subset pairwise TM, lower = more diverse; colour marks REPA *vs.* baseline at the same step and bin: red / green = more concentrated/more diverse, $|\Delta|$<0.05 uncoloured. n≤256, single-seed; *900K steps.)

but rather fails to grow out of the early-training regime that the baseline leaves behind. Notably, the AFDB-MPNN configuration routes *away* from β-rich folds rather than toward them. It sheds designable sheet content where GearNet adds it (the E%des column of Table 6.6: −0.04 against +0.04). This lets it escape the concentration trap entirely, and shows the behaviour is modulated by both dataset and encoder.

Nevertheless, REPA's whole-set gains persist deep into training. At 1.0M the strongest variant on each dataset keeps its fidelity and coverage lead — REPA-MPNN-L9 on PDB (FID −81, fold-score +0.29) and REPA-GN-L4 on AFDB (−228, +3.68) — even as their distinct designable-fold counts fall below baseline (−51 and −39 clusters).

Does alignment outlive its purpose? Given the early acceleration in generation quality induced by REPA, and the later-stage fold concentration we observe, it is possible that the best approach may be to stop alignment after early training. A similar arc is reported in image diffusion literature, where REPA is reported to help early and is best switched off after early training [13]. We do not test such a schedule and make no argument of this form. However, if this framework holds, an early stop may let REPA diversify like the baseline while keeping the whole-set gains.

6.2.5 REPA's gains hold across sampler noise levels

REPA's gains reproduce across the sampler's noise range. Proteina samples with an SDE whose noise scale γ trades designability for diversity. REPA and the baseline travel this trade-off in parallel — REPA shifts the levels, not the shape. Across the full sweep, REPA's designability and fidelity gains hold (Table 6.8). The γ=0.45 setting we report elsewhere is therefore representative, not a favourable pick.

REPA also raises the deterministic-sampling floor. With the sampler noise switched

Δ (REPA − baseline)	ODE	$\gamma = 0$	$\gamma = 0.35$	$\gamma = 0.45$	$\gamma = 0.5$	$\gamma = 1.0$
PDB — REPA L9-MPNN vs Baseline (step-matched cells).						
Des ↑	+0.06	+0.07	+0.16	+0.18	+0.15	+0.04
FID ↓	−22	−1225	−78	−77	−89	−131
fJSD-A ↓	−0.13	+0.18	+0.21	−0.06	+0.07	−0.30
ssJSD2D ↓	−0.11	+0.08	−0.01	−0.03	−0.00	−0.10
#Clust ↑	+9	+8	−7	+4	+12	+2
AFDB — REPA L4-GearNet vs Baseline (step-matched cells).						
Des ↑	+0.16	+0.08	+0.07	+0.06	+0.06	+0.08
FID ↓	−102	−45	−133	−144	−132	−97
fJSD-A ↓	−0.31	+0.28	−0.94	−1.06	−0.92	−0.24
ssJSD2D ↓	−0.04	−0.19	−0.13	−0.13	−0.10	−0.03
#Clust ↑	+32	−15	−27	−23	−21	+10

Table 6.8: **REPA's gains are sampler-invariant in the middle band ($\gamma \in [0.35, 0.5]$) and largely survive at the extremes — a training-time effect that composes with the inference-time noise dial.** The large −1225 FID gain at γ=0 reflects the PDB baseline mode-collapsing at deterministic sampling. (Mean Δ = REPA − baseline across step-matched cells per noise level, for the best PDB and AFDB variants; green / red = improvement/regression; negative is a win on lower-is-better metrics.)

off (the ODE limit), the baseline's designability bottoms out — 0.04 on PDB and 0.12 on AFDB. Every learned REPA variant lifts the AFDB floor, by +0.10 to +0.28 across checkpoints; on PDB the learned encoders lift it (up to +0.16) while the random control does not (−0.02). As deterministic sampling is the cheapest and most reproducible regime, this is among the more practically useful of REPA's gains.

6.2.6 What we do not claim

Several tempting claims do not survive our own data, and we flag them. REPA does not reliably improve novelty — the signal is too noisy to call. It does not "preserve β" in general; the direction is encoder×dataset conditional, and MPNN-AFDB reverses it. It does not strictly *require* a learned encoder at every scale — at $n \leq 128$ a random encoder helps nearly as much (§6.2.7). Its designability gains are read through a proxy (ProteinMPNN→ESMFold) that shares lineage with AFDB's AlphaFold2, so we never compare designability across datasets without that caveat. We claim co-movement, not formal mediation, between representation and generation. And we retract an earlier "distribution collapse at γ=1" reading: with the full encoder grid it was a single-cell artefact.

6.2.7 Robustness across scale

At smaller scale the learned-versus-random gap narrows, so we anchor headline claims at $n \leq 256$. At $n \leq 128$ the random control helps nearly as much on representation and whole-set metrics. The direction of every gain is consistent across scale, but the margin of learned over random shrinks, so we read $n \leq 128$ as a scale-robustness check

rather than a headline regime. Training-dynamics ablations (λ, batch size, projector depth) are reported in Appendix A; length extrapolation — generating proteins longer than the training crop — is the natural next test and remains future work.

6.2.8 Summary

REPA does real mechanistic work on Proteina — but as a family of encoder-routed interventions, not a single effect. It installs structural representations the flow-matching loss never learns, as a persistent gap; it pulls the trunk's geometry onto the encoder's, generalising across encoders in the Platonic sense; and it accelerates generation, durably on synthetic data and transiently on experimental data. The representation and generation gains are encoder-matched, strong evidence that the one runs through the other. The gain is generic on whole-set distribution, encoder-specific on quality, and carries a designable-diversity trade-off that develops with training and splits into a geometric-bias clustering component and a learned-feature directional one. Which encoder you align to chooses which of these you get. We carry this — and the contrast with the small-molecule study of Chapter 5 — into Chapter 7.

Chapter 7

Conclusions

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Appendix A

Technical details

A.1 Dataset composition and split overlap

Table A.1 summarises the two datasets used in Chapter 6. Both training sets are α/β-dominated and span a substantial fraction of CATH architectures (44.2% PDB, 61.8% AFDB), which is what makes the per-chain CATH fold probe meaningful. Neither random split is sequence-clustered: 98.3% of PDB and 92.2% of AFDB validation chains have a $\geq$30%-identity homolog in their own training set, so absolute in-house scores are inflated and we read rank-order and Δ-from-baseline instead. The cross-database overlap is far lower — only 62.1% of AFDB validation chains have a $\geq$30%-identity hit in PDB training — so a homology-filtered AFDB set serves as a cleaner probe set for PDB-trained models (Chapter 3).

A.2 Choice of CATH fold-probe level

We report CATH *architecture* (CATH-A) as the headline fold-probe level throughout Chapter 6. Table A.2 justifies that choice: Class is too coarse (four buckets, one already near-saturated), Topology too finely-bucketed to probe reliably, and Architecture sits between — discriminative but learnable.

Property	**PDB**	**AFDB (Swiss-Prot)**
Origin	Experimental (X-ray, cryo-EM)	Predicted (AlphaFold2, v6)
Residue-length crop n	$\leq$ 256 residues	
Train / val / test chains	273,771 / 3,190 / 323	229,670 / 4,521 / 235
CATH coverage (train)	44.2%	61.8%
Val$\leftrightarrow$train, $\geq$30% id.	98.3%	92.2%
AFDB-val$\leftrightarrow$PDB-train, $\geq$30% id.	62.1%	

Table A.1: Properties of the two datasets we study, at the headline $n\leq$256 regime. Sequence identity is computed with MMseqs2 [28] (`easy-search`, $\geq$30% identity over $\geq$80% alignment coverage).

CATH level	#classes	baseline probe acc.	verdict
C (class)	4	0.733	one class (α/β) is 55% of the data; coarse, already near-saturated — *gameable*
A (architecture)	∼40	0.407	balanced and discriminative, with headroom to move — **our headline fold metric**
T (topology)	∼1400	0.301	long-tailed (top fold ∼18%; only ∼89 classes populated enough to probe); per-class accuracy is unreliable — *too many buckets*

Table A.2: **We read fold-level representation quality off CATH *architecture* — Class is too coarse to discriminate, Topology too finely-bucketed to probe reliably.** Class has 4 buckets (one at 55%, baseline already 0.73); Topology has ∼1400 imbalanced buckets; Architecture sits between — enough classes to be meaningful, enough examples per class to be learnable — so CATH-A is our headline fold metric (C and T are reported in the appendix). ("Baseline probe acc." is a linear probe on the un-aligned trunk.)

A.3 Leakage-controlled probe evaluation

The random train/val split (§A.1) is not sequence-clustered, so probe scores are exposed to two distinct leakages. *Model leakage* is asymmetric: a model trained on the leaky split has effectively seen near-duplicates of the evaluation proteins, an advantage a differently-curated model (such as an externally-pretrained checkpoint) does not share. *Probe leakage* is symmetric: a probe fit on training chains near-identical to its evaluation set can shortcut by matching sequences rather than learning generalisable structure, inflating every model's absolute score.

We control these by filtering the evaluation. A *probe-clean* split removes probe leakage alone — the probe's training chains are filtered to have no ≥30% sequence-identity hit to the evaluation set, which is otherwise the full validation set. A *blinded* split removes both — the evaluation proteins are held to under 30% identity to all training data (here, a cross-database set of AFDB chains with no such hit in either the PDB or AFDB training pool, ∼325 proteins). We use the blinded split for the per-residue probes (inverse folding, dihedral), where the ∼325 proteins still yield ∼43k residues — full statistical power. The per-chain CATH probe is different: fold classification needs many distinct chains spread across populated fold classes, and the blinded slice is too small and too class-sparse for this, so we relax it to the probe-clean split on the fuller validation set (∼3,190 chains). Comparing the splits also decomposes the leakage — the dirty-to-probe-clean drop isolates probe leakage and the probe-clean-to-blinded drop isolates model leakage — and the REPA-over-baseline gap survives in every regime.

A.4 The ssJSD-2D secondary-structure metric

Proteina's secondary-structure metric compares the *mean* helix and strand content of a generated set against a reference. This collapses each set to a single point, so it cannot see variance, bimodality, or tails — a model that matches the average composition while collapsing its spread scores the same as one that reproduces it. ssJSD-2D removes this blind spot by comparing the full joint distribution.

For each backbone we annotate every residue as helix, strand, or coil with a Cα-only secondary-structure assignment (P-SEA, as implemented in Biotite) and record the fraction of residues in helix (f_H) and in strand (f_E). A set of N backbones is then a cloud of N points in the (f_H, f_E) plane. We bin that plane into a 5×5 grid over $[0, 1]^2$, form normalised two-dimensional histograms for the generated and reference sets, and report the Jensen–Shannon divergence between them (lower is better). Binning the joint distribution, rather than comparing marginal means, lets the metric reward a model for reproducing the *shape* of the secondary-structure distribution and penalise mode collapse onto, say, all-α backbones.

A.5 Placeholder section

TODO.